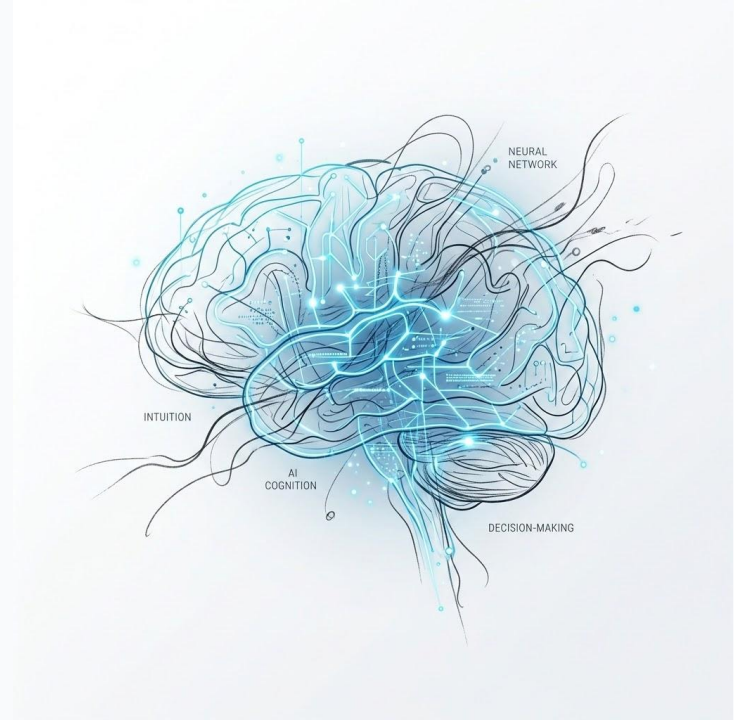


Data

What can a computer “see,”
and what has to happen
before the world becomes
computable?

SYS 478. Fall 2026. Prof. Sarah Van Wart



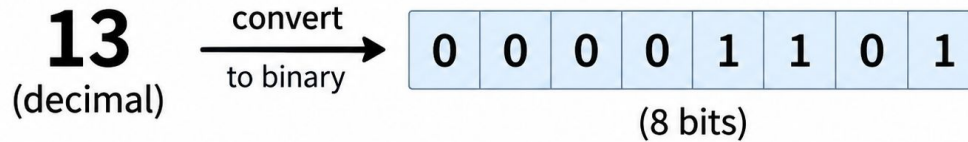
1. Binary Representation

Why binary?

- Computers encode data using physical switches (usually transistors).
- We conventionally represent those states as 0 and 1 (on or off), also known as binary notation.
- These binary (on / off) switches can be combined in different ways to represent different things.
- Binary is not the information itself – it only becomes information when operating inside of an encoding system.

Numbers: Decimal and Binary

Computers represent numbers using binary (base 2).



Place value (in decimal)	128	64	32	16	8	4	2	1
Bits	0	0	0	0	1	1	0	1

$$00001101_2 = 13_{10}$$

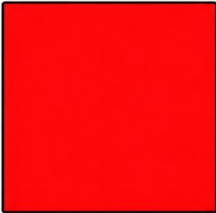
The value comes from how we interpret the bits.

[Worksheet](#)

But what does a number like
01 0001
actually mean?

Colors: RGB Values

A computer screen represents color using numbers for red, green, and blue (RGB).



R	255
G	0
B	0



1 1 1 1 1 1 1 1

Red
(8 bits)

0 0 0 0 0 0 0 0

Green
(8 bits)

0 0 0 0 0 0 0 0

Blue
(8 bits)

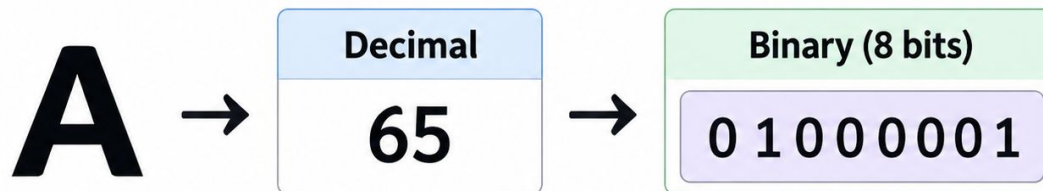
To represent color computationally,
we **create a system**
for turning color into numbers.

[Demo](#)

What information about "redness" does RGB preserve and ignore?

Text: ASCII Encoding

Computers represent text using a character encoding such as ASCII, where each character has a number.



Character	Decimal	Binary (8 bits)
A	65	01000001
B	66	01000010
a	97	01100001

Bits do not carry meaning by themselves.
They mean 'A' because of the encoding scheme.

[Demo](#)

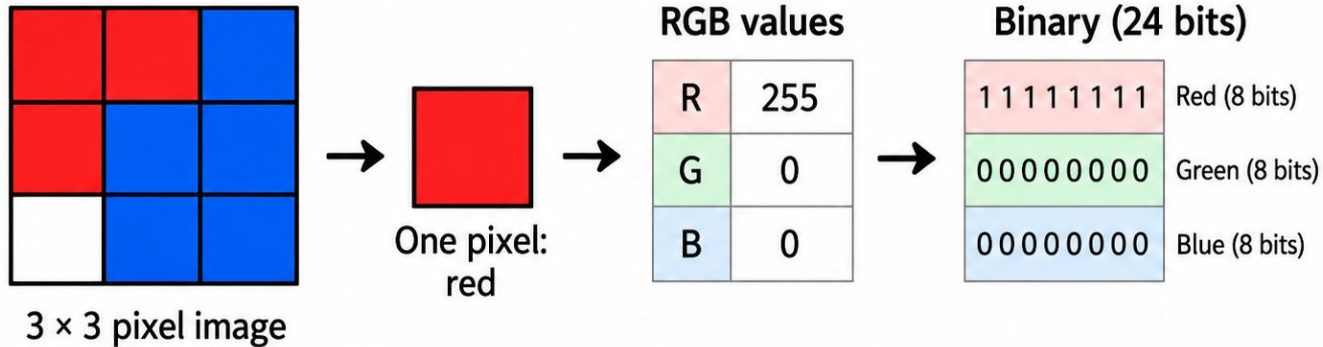
MIDI Notes

Note	-1	0	1	2	3	4	5	6	7	8	9
C	0	12	24	36	48	60	72	84	96	108	120
C#	1	13	25	37	49	61	73	85	97	109	121
D	2	14	26	38	50	62	74	86	98	110	122
D#	3	15	27	39	51	63	75	87	99	111	123
E	4	16	28	40	52	64	76	88	100	112	124
F	5	17	29	41	53	65	77	89	101	113	125
F#	6	18	30	42	54	66	78	90	102	114	126
G	7	19	31	43	55	67	79	91	103	115	127
G#	8	20	32	44	56	68	80	92	104	116	
A	9	21	33	45	57	69	81	93	105	117	
A#	10	22	34	46	58	70	82	94	106	118	
B	11	23	35	47	59	71	83	95	107	119	

2. Representing something more complicated

Images: Bitmaps

A digital image is a grid of pixels, and each pixel stores numerical color values.

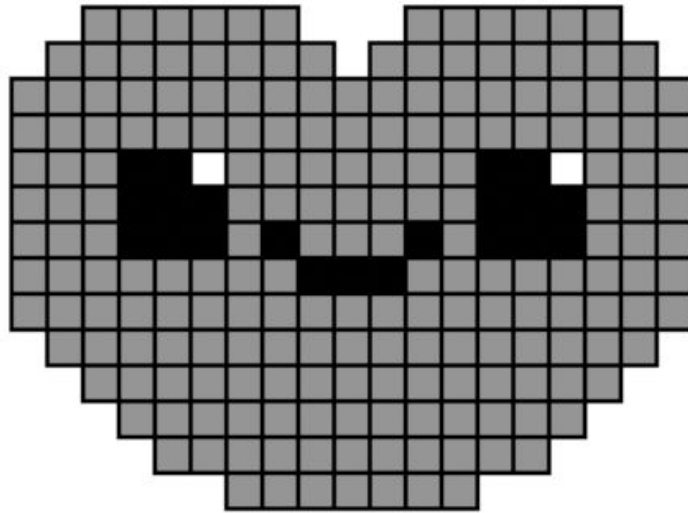


An image is a collection of numbers representing color at each pixel.

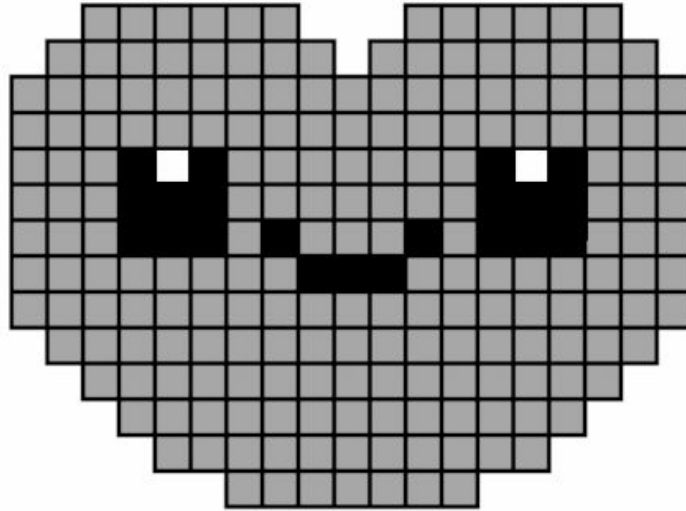
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150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150
150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150
150	150	150	255	0	0	150	150	150	150	150	150	150	255	0	0	150	150	150
150	150	150	0	0	0	150	150	150	150	150	150	150	150	0	0	0	150	150
150	150	150	0	0	0	150	0	150	150	150	0	150	0	0	0	150	150	150
150	150	150	150	150	150	150	150	0	0	0	150	150	150	150	150	150	150	150
150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150
255	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	255
255	255	150	150	150	150	150	150	150	150	150	150	150	150	150	150	150	255	255
255	255	255	150	150	150	150	150	150	150	150	150	150	150	150	150	255	255	255
255	255	255	255	150	150	150	150	150	150	150	150	150	150	150	255	255	255	255
255	255	255	255	255	255	150	150	150	150	150	150	150	255	255	255	255	255	255

Representing
an image as
binary

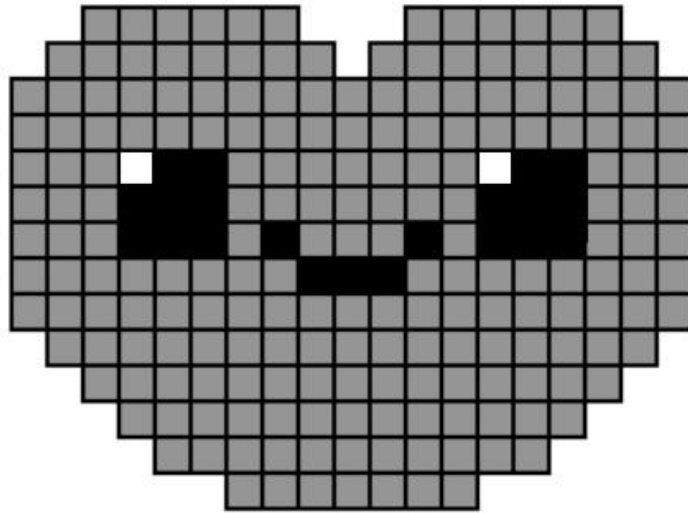
What changes?



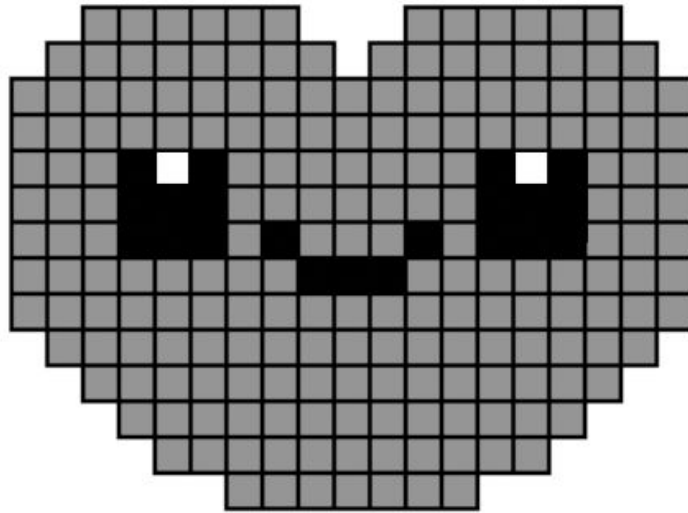
What changes?



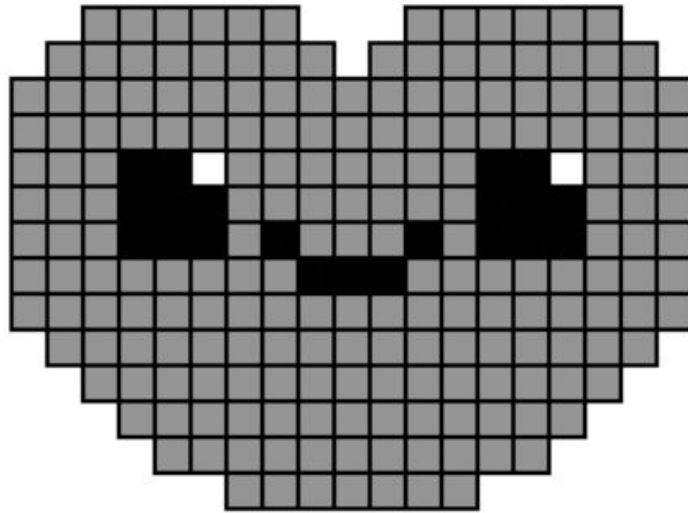
What changes?



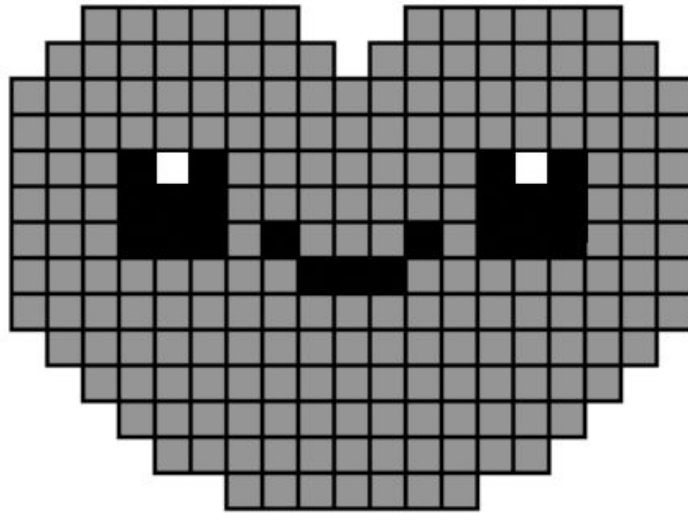
What changes?



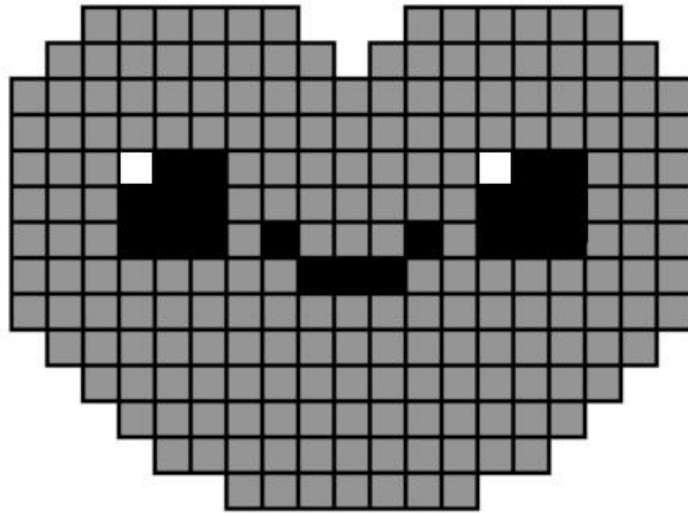
What changes?



What changes?



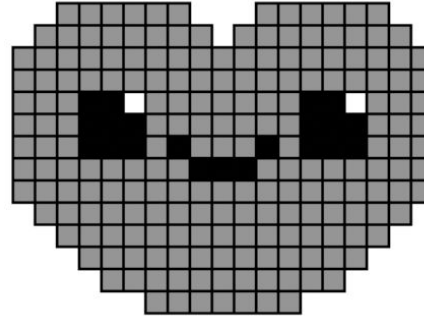
What changes?



Humans v. Computers

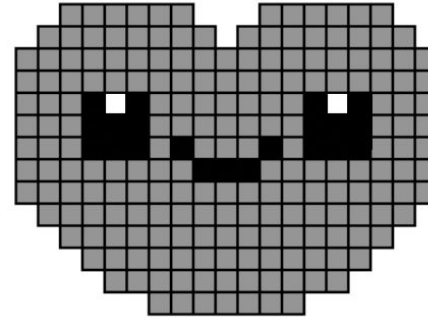
- How different are they to you? What meaning is conveyed?
- How different are these images to the computer?

255 = white
200 = light gray
150 = gray
100 = dark gray
0 = black



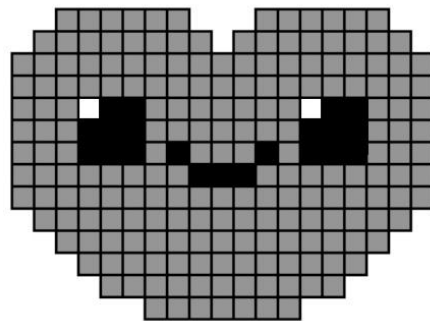
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)
```

255 = white
200 = light gray
150 = gray
100 = dark gray
0 = black



```
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)
```

255 = white
200 = light gray
150 = gray
100 = dark gray
0 = black



```
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)
```

3. The physical world has
to be sampled

Stuart Hall: Cultural & Media Theorist (1932-2014)

Undergraduate work at Jamaica College

Rhodes Scholar; studied at Oxford. Went onto the faculty of the University of Birmingham and later the Open University

Known for representation theory: the production of representations and their interpretation are dynamic and subjective.

Representations in the media are a way to construct and perpetuate ideas about the world from the dominant perspective.

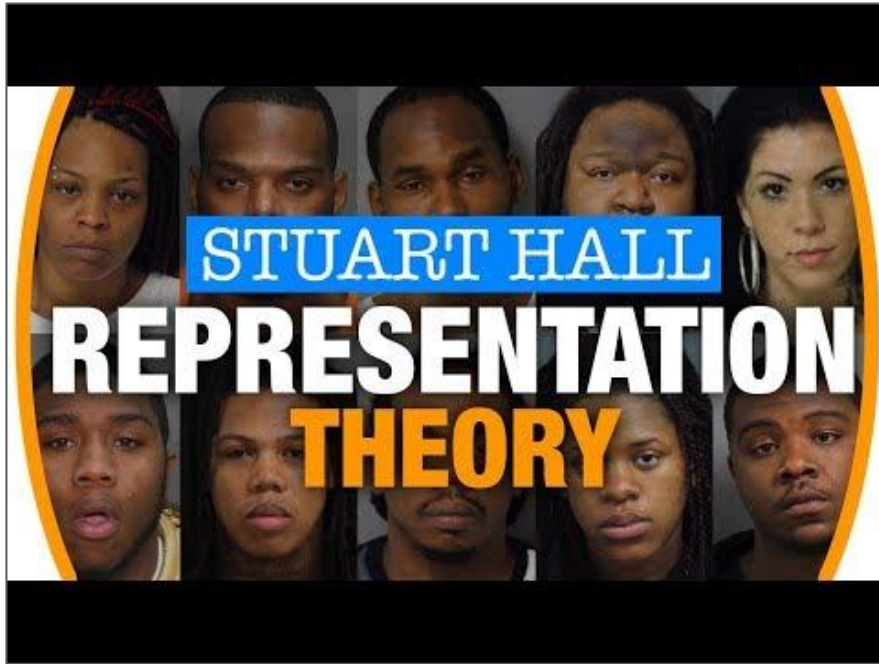


Stuart Hall: Photography

A critical lens on photography as an authoritative source of truth:
When using photographs to make sense of the world, one must critically examine:

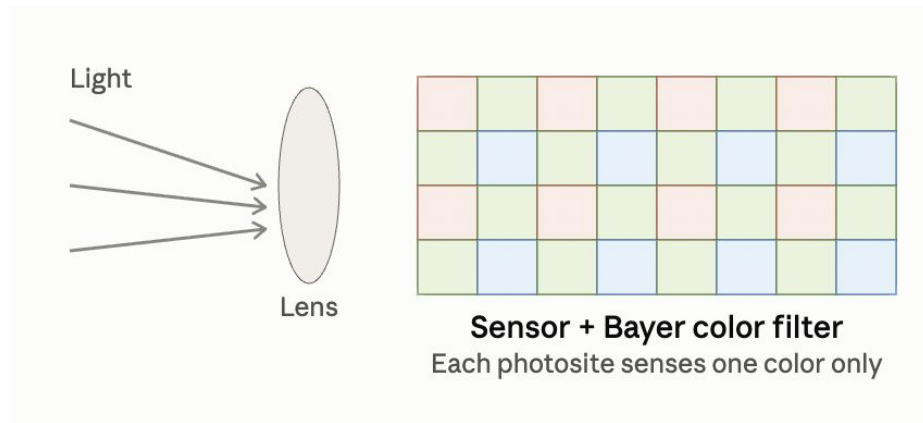
1. who produced these images
2. what purpose they serve
3. how they further their agenda (e.g., what has been deliberately included and excluded in the frame).

Stuart Hall: Representation & Reception Theory



Photography & Video

- **Light comes in:** the lens focuses light onto the sensor
- **Millions of tiny buckets (photodiodes) collect light:** each one only “sees” red, green, or blue, because of a filter sitting on top of it.
- **The camera’s chip figures out how to assign a color to each dot:** every dot now has a red, green, and blue value (the bitmap).



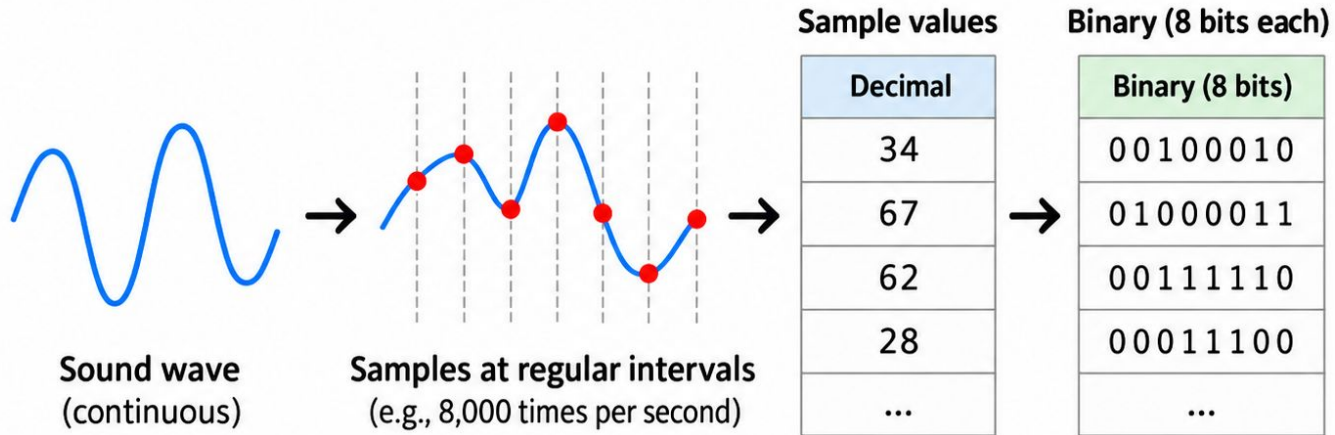
What happens when a computer "hears" you?

Your thoughts here...

The computer never receives *your voice*.
It receives a representation of it.

Sound: Sampling a Wave

A sound wave is a continuous signal. A computer takes samples at regular intervals and stores each sample as a number.



A digital recording is a sequence of numbers representing the sound wave.

Analog vs. Digital

Analog → continuously varying signal

Digital → discrete measurements represented as numbers

What gets lost? Does it matter?

Representation $\neq$ Reality

A representation is not the thing itself.

World	Representation
sound	samples
image	pixels
text	character codes / tokens
person	record
student	selected variables

Every representation
preserves some
information and leaves
other information out.

Activity

Tree Debrief

Your thoughts here...

Mortgage Screening

Your thoughts here...

Representation is a Choice

Before an algorithm can make a judgment about the world, someone has already made decisions about how that world will be represented.

Representation → What becomes data?

- What gets measured?
- What gets omitted?
- What gets simplified?
- Who decided how to encode it?

Human Subjectivity & Context Matter!

- With sound, we choose how many times to sample a waveform.
- With people, institutions also “sample” reality – they record some things and not others.

Some Famous Examples

From the Order of things (Foucault, 1970)

“This book first arose out of a passage in [Jorge Luis] Borges, out of the laughter that shattered, as I read the passage...[which] quotes a ‘certain Chinese encyclopaedia’ in which it is written that ‘animals are divided into: (a) belonging to the Emperor, (b) embalmed, (c) tame, (d) suckling pigs, (e) sirens, (f) fabulous, (g) stray dogs, (h) included in the present classification, (i) frenzied, (j) innumerable, (k) drawn with a very fine camel hair brush, (l) et cetera, (m) having just broken the water pitcher, (n) that from a long way off look like flies’.

In the wonderment of this taxonomy, the thing we apprehend in one great leap, the thing that, by means of the fable, is demonstrated as the exotic charm of another system of thought, is the limitation of our own, the stark impossibility of thinking that.”

The Dewey Decimal System

Started in 1876, the DDC is the most widely used classification system in the world, especially in public libraries:

000 Generalities

100 Philosophy & psychology

200 Religion

300 Social sciences

400 Language

500 Natural sciences & mathematics

600 Technology (Applied sciences)

700 The arts

800 Literature & rhetoric

900 Geography & history

The Dewey Decimal System

Started in 1876, the DDC is the most widely used classification system in the world, especially in public libraries:

200 Religion

210 Natural theology

220 Bible

230 Christian theology

240 Christian moral & devotional theology

250 Christian orders & local church

260 Christian social theology

270 Christian church history

280 Christian sects & denominations

290 Other religions

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F -- HISTORY: AMERICA
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H -- SOCIAL SCIENCE
J -- POLITICAL SCIENCE
K -- LAW
L -- EDUCATION
M -- MUSIC
N -- FINE ARTS
P -- LANGUAGE AND LITERATURE
Q -- SCIENCE
R -- MEDICINE
S -- AGRICULTURE
T -- TECHNOLOGY
U -- MILITARY SCIENCE
V -- NAVAL SCIENCE
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